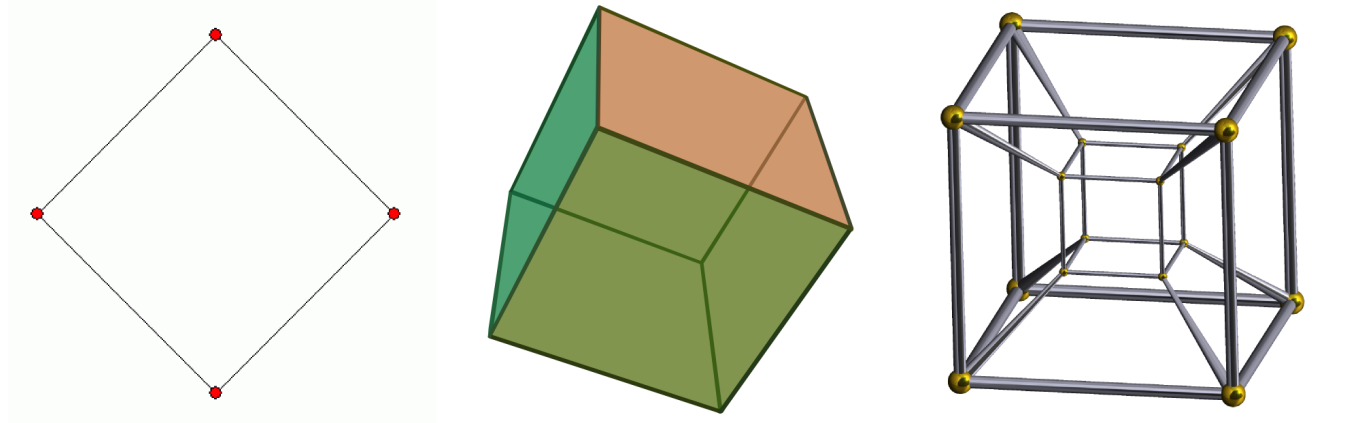


Lab-5 Ex6 - High Dimension Dot-Product

This is an extension to the previous array exercise in **Lab-4 Ex4** on calculating the *dot product* of two 3-element vectors.

Instead of limiting ourselves to 3-dimension, let's try n -dimension!



A square (2D), cube (3D) and tesseract (4D) visualization. Source: <https://en.wikipedia.org/wiki/Dimension#/media/File:Squarecubeesseract.png>

Write a C program to ask the user for an input integer n (which is assumed to be positive and **not more than 100**). Then ask for two vectors v and w . Each vector has n double-type values that are stored in an n -element array, e.g., $v[0]$, $v[1]$, ..., $v[n-1]$ and $w[0]$, $w[1]$, ..., $w[n-1]$ for vectors v and w respectively. Calculate and print the *dot product* (scalar product) of v and w , in 2 decimal places.

User-input numbers are ***bold and italic*** in the following sample runs.

Sample run #1:

```
Input n: 3
Enter vector v: -4 0 2.5
Enter vector w: 3 -9.876 5
Dot product v.w = 0.50
```

Sample run #2:

```
Input n: 10
Enter vector v: 1 2 3 4 5 6 7 8 9 10
Enter vector w: 10 0 0 0 0 0 0 0 0 9
Dot product v.w = 100.00
```
